

Yusuf Naeem Abd El-Hakim

Software Engineer | Systems Design & Applied Security

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SUMMARY

Software engineer who takes systems from a blank schema to production without supervision. Three years designing REST APIs, relational data models, and access-control layers built to hold under concurrent load, alongside two self-directed SaaS products taken solo from idea to public launch. Owns what comes after launch too: reliability, performance, and security hardening, with offensive-security training in the OWASP Top 10 shaping how systems are threat-modelled before they ship rather than patched afterwards. Equally comfortable on the commercial side, gathering requirements across business units, running cost-benefit analysis for executive decisions, and redesigning the processes the software is meant to serve. Now formalising that practice through graduate research in distributed systems and software system design.

EDUCATION

B.Sc. Computer Science | Software Engineering Major Ahram Canadian University (ACU) · GPA 3.36 / 4.00 Sep 2019 – Jul 2023

Activities: ACU Competitive Programming Community · CyberTalents

Volunteering: Peer tutoring for classmates in data structures, algorithms, and software project management, over more than a year.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL

Backend: Node.js, Express.js, REST API design, PostgreSQL, Supabase, SQLite, multi-tenancy & row-level security, OAuth 2.0 / JWT authentication

AI / LLM: Gemini API integration, semantic search, prompt engineering

Frontend: React, Next.js, TanStack Query, Tailwind CSS, Three.js, D3, HTML, CSS

Infrastructure: Docker, Nginx, Vercel, AWS (S3, EC2), Liquibase, CI/CD, caching & performance tuning

Practices: Git, Agile / Scrum, TDD, BDD (Cucumber), Postman, Swagger, Jira

Also worked with: Java / Spring Boot, Django REST Framework, Power BI, OWASP Top 10, Burp Suite

PROFESSIONAL EXPERIENCE

Software Engineer | EPPS Egyptian Packaging & Plastic Systems, Cairo · Hybrid · Full-time since Jun 2024 Apr 2024 – Present

- Built and now operate an internal B2B lead pipeline platform replacing spreadsheet tracking across four business entities, holding roughly 2,000 records for the 8 to 10 staff running marketing capture through to closed sale.
- Designed a full audit trail and an atomic identifier generator that keeps records conflict-free under concurrent writes.
- Resolved a Marketing and Sales data-ownership conflict with field-level, server-side access control that strips restricted fields from API payloads before they reach the client.
- Built SC 3D Optimizer, applying a 3D bin-packing heuristic to container load planning under door-clearance constraints, cutting planning time roughly fourfold and improving space utilisation (React, Three.js).
- Modelled the data and lifecycle workflows for the company's IT asset management platform, tracking roughly 3,000 assets with full check-out and check-in history.
- Own reliability and security across 25+ domains and 7 production sites through caching, automated backups, database cleanup, and web server hardening.
- Co-architected IT governance, standard operating procedures, and Active Directory infrastructure for 90+ users.

Offensive Security Engineer | WE INNOVATE Cairo, Egypt · Hybrid · Internship Sep 2024 – Dec 2024

- Exploited web, mobile, and infrastructure vulnerabilities against the OWASP Top 10, documenting every finding with a reproducible proof-of-concept and a remediation path (Burp Suite, Metasploit, Wireshark).
- Ran Active Directory and network penetration tests, mapping privilege-escalation and lateral-movement paths across the target environment.
- Wrote Python tooling to automate recurring reconnaissance and vulnerability-assessment workflows, beyond the defined scope of the internship.

Software Engineer | HomeBay Furniture Ontario, Canada · Remote Nov 2023 – Oct 2024

- Automated product catalogue and asset synchronisation through the Google Sheets API, removing a recurring manual cataloguing task from the team's weekly cycle (Python).
- Extended WooCommerce product-management and checkout functionality, and applied caching, database indexing, and image optimisation across the storefront.

Software Engineer | Kites Software Cairo, Egypt · Hybrid · Internship Aug 2023 – Nov 2023

- Built and shipped RESTful API endpoints test-first for Egypt's first SaaS ERP for import and distribution, writing integration tests in BDD style with Cucumber (Java, Spring Boot, PostgreSQL).
- Diagnosed and fixed QA-reported defects inside a full Agile and Scrum cycle.
- Authored onboarding documentation that the team adopted for subsequent intern cohorts.

- Led the company's first digital-transformation initiative, replacing informal, untracked coordination with a centralised system.
- Built a CRM and operations hub in Notion from scratch, giving site engineers, architects, and supervisors a single platform for task assignment and progress tracking.
- Redesigned cross-functional workflows between field and office teams, and documented the new process as a standard other staff could follow.

Freelance Full-Stack Developer Remote · Self-employed

Jan 2022 – Jun 2022

- Built and shipped websites and online storefronts for small businesses end to end, from static frontend (HTML, CSS, vanilla JavaScript) through backend logic and data (Django, Django REST Framework).
- Worked directly with each business owner to scope requirements and turn them into a working site without a team or supervision.
- Implemented storefront features, including product catalogues and inquiry forms, backed by Django REST Framework APIs.

SELECTED PROJECTS & PRODUCTS

Commit | Curriculum Tracker for Self-Taught Developerscommit.binhakim.dev

Sole Designer and Engineer · Launched May 2026 · Node.js, PostgreSQL, SQLite, Resend

- Designed a git-conceptualised learning model mapping study sessions to commits, streaks to contribution graphs, and course completions to merges, turning unstructured self-study into a measurable engagement loop.
- Built a database-agnostic data access layer allowing a single-line swap between local SQLite and production PostgreSQL, eliminating environment drift and reducing setup to one command.
- Implemented an ETA estimation engine projecting completion dates from observed study pace, alongside roadmap visualisation, XP milestones, and automated daily digests (Resend).
- Took the product from personal tool to public launch: scoped requirements, designed the schema, implemented the API, and now operate it in production.

Orbit | Professional Relationship Operating Systembyorbit.io

Sole Designer and Engineer · Live, in active development · Next.js, TypeScript, Supabase (PostgreSQL), Gemini, Stripe

- Enforced tenant isolation at the database layer with PostgreSQL row-level security, so isolation holds even when application code is wrong.
- Implemented Google OAuth and magic-link authentication, and Stripe Free/Pro billing with usage metering.
- Integrated the Gemini API to power semantic search across a user's connections, and built a D3 force-directed graph for exploring network structure.
- Shipped full English and Arabic right-to-left localisation, packaged as an installable Progressive Web App.

iREAD | Reading Platform and Digital Bookstore

2023

Graduation Project, Ahram Canadian University · Supervised by Dr. Sherif El-Shafey, Head of Software Engineering · Flutter, React, REST API

- Built a reading platform combining a catalogue with reviews and reading history, a storefront for digital book purchases, and a community layer for readers to exchange insights.
- Designed a REST API serving both the Flutter mobile client and the React web client, keeping a single source of business logic across the two platforms.

PUBLICATIONS

Technical Articles | Medium (@binhakim)

Aug 2025 – Oct 2025

- Published articles on backend engineering, database design, and web security, translating technical material for practitioner and general audiences.

CERTIFICATIONS

- CS50: Introduction to Computer Science, HarvardX
- Web Development and .NET Development Summer Training, Information Technology Institute (ITI)
- Offensive Security Engineering, Certificate of Completion, WE INNOVATE
- Forward Program, McKinsey.org

HONOURS & ACHIEVEMENTS

- Honourable Mention, Egyptian Collegiate Programming Contest (ECPC), 2021

LANGUAGES

Arabic (native or bilingual) · English (professional working) · German (limited working)